Journal of Ethnopharmacology 302 (2023) 115892

Contents lists available at ScienceDirect

Journal of Ethnopharmacology

journal homepage: www.elsevier.com/locate/jethpharm

Si–Ni-SAN ameliorates obesity through AKT/AMPK/HSL pathway-mediated lipolysis: Network pharmacology and experimental validation

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ARTICLE INFO

Keywords:
Si-Ni-San
Obesity
Network pharmacology
Mechanism

ABSTRACT

Ethnopharmacological relevance: Si–Ni-San (SNS) is a famous Chinese herbal formula used in China for thousands of years. It has clinical effects on a variety of lipid metabolism disorders, but the ameliorating effects of SNS on obesity and underlying mechanisms remained poorly elucidated.
Aim of the study: This study aims to explore the therapeutic effect and mechanism of SNS on obesity from multiple perspectives in vitro and in vivo.
Materials and methods: The high-fat diet (HFD)-induced obesity mouse model was established to evaluate the effect of SNS. Then network pharmacologic methods were performed to predict underlying mechanisms, and the core pathways were verified in animal and cell studies.
Results: Our results demonstrated that SNS significantly reduced body weight, body fat content, white adipose tissue (WAT) expansion in obese mice, and lipid accumulation in primary mouse embryonic fibroblasts (MEFs) cells. Network pharmacologic analysis identified 66 potential therapeutic targets, and Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis of these genes revealed that the most important signaling pathway includes AMP-activated protein kinase (AMPK) signaling pathway, regulation of lipolysis in adipocytes, lipid and atherosclerosis. Western blot assay confirmed that SNS activated hormone-sensitive triglyceride lipase (HSL) and adipose triglyceride lipase (ATGL) activity and promoted lipolysis through AMPK signaling pathway.
Conclusion: The results confirmed that SNS improves lipid accumulation through AKT/AMPK/HSL axis mediated lipolysis, which opens a new option for clinical treatment of obesity and associated complications.

1. Introduction

Obesity is defined as a condition when someone's body mass index (BMI), a weight divided by the square of a person's height, is greater than or equal to 30 (Bluher, 2019). The survey found that the obesity prevalence of 25.5% among adults aged 18–34 years in the United States (Watson et al., 2022). In China, the prevalence of overweight and obesity among adults aged 18 and above was 34.3 percent and 16.4 percent, respectively (Pan et al., 2021). The rapid increase in overweight and obesity has become a worldwide public health problem that cannot be ignored. Obesity contributes to the incidence of diseases such as type 2 diabetes (Jiang et al., 2022; Nestle, 2022) and cardiovascular disease (Virani et al., 2021) and affects cancer incidence and mortality (Rock et al., 2022). Because obesity complicates the management of multiple diseases, the development of anti-obesity medications (AOMs) has attracted much attention. Excessive energy intake and lipogenesis lead to obesity. On the contrary, lipolysis and oxidation of energy can

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https://doi.org/10.1016/j.jep.2022.115892
Received 28 August 2022; Received in revised form 28 October 2022; Accepted 29 October 2022
Available online 2 November 2022
0378-8741/

Abbreviations

SNS	Si–Ni-San
TCM	traditional Chinese medicine;
MEFs	primary mouse embryonic fibroblasts
KEGG	Kyoto Encyclopedia of Genes and Genomes
AMPK	AMP-activated protein kinase
HSL	hormone-sensitive triglyceride lipase
ATGL	adipose triglyceride lipase
AKT	protein kinase B
AOMs	anti-obesity medications
WAT	white adipose tissue
CH	Bupleuri Radix
BS	Paeoniae Radix Alba
ZS	Aurantii Fructus Immaturus
GC	Glycyrrhizae radix et Rhizoma Praeparata Cum Melle
NAFLD	non-alcoholic fatty liver disease
HFD	high-fat diet;
H&E	hematoxylin and eosin
CM	complete medium
PPI	protein-protein interaction
GO	Gene Ontology
TCMSP	traditional medicine system pharmacology technology platform
TG	triglycerides
TAG	triacylglycerol
NEFA	nonesterified free fatty acids

effectively inhibit fat accumulation (Hafidi et al., 2019). To achieve optimal weight loss, it is obvious that pharmacological treatment must target both energy intake and expenditure. Under these conditions, several AOMs are available, the most commonly used of which are orlistat, phentermine-topiramate, semaglutide, Naltrexone-bupropion, liraglutide, etc., and control the energy balance through peripheral or central pathways (Shi et al., 2022). Despite the favorable results in reducing obesity, most of these drugs show limited efficacy, patient heterogeneity, and significant side effects such as oil spots and diarrhea (Muller et al., 2022). Until recently, safe and effective long-term pharmacotherapy remained elusive (Muller et al., 2018).

Unlike the current single-target drug development strategy, traditional Chinese medicine (TCM) prescriptions have the characteristics of multi-pathway, multi-target, and high safety, which is a rich source for the discovery of anti-obesity drugs. The understanding and treatment of obesity in TCM have a long history, as early as the ancient book "Nei Jing" has been recorded (Zhang, C.H. et al., 2021a). According to the TCM theory, spleen deficiency and dampness obstruction (Pixu Shizu) and liver depression and Qi stagnation (Ganyu Qizhi) are the most common typing of syndrome differentiation of obesity, which its primary therapy are strengthening the spleen and Supplementing qi (Jianpi Yiqi), and dispelling melancholy by soothing liver and relieving depression (Shugan Jieyu). Si–Ni-San (SNS) has been used in China for thousands of years and was first recorded in Shang Han Lun by Zhang Zhongjing, a famous doctor in the Han Dynasty. Its traditional use is to maintain the functional balance between the liver and the spleen, and TCM uses its functions of soothing the liver and relieving depression (Shugan Jieyu), strengthening the spleen and Supplementing qi (Jianpi Yiqi) to treat a variety of diseases caused by obesity (Wei et al., 2021; Zhu et al., 2019).SNS consists of Bupleuri Radix (*Bupleurum chinense* DC., CH), Paeoniae Radix Alba (*Paeonia lactiflora* Pall., BS), Aurantii Fructus Immaturus (*Citrus trifoliata* L., ZS), and Glycyrrhizae radix et Rhizoma Praeparata Cum Melle (*Glycyrrhiza uralensis* Fisch., GC) in the ratio of 1:1:1:1 (Cai et al., 2021). Significant reductions in serum triglycerides (TG) and cholesterol were found in patients in the course of SNS treatment for NAFLD. Furthermore, Wei et al. found that it significantly reduced body weight and serum TG content in rats induced by a high-fat diet (HFD) (Wei et al., 2021). Similar to other prescriptions, multiple drugs and components in SNS interact with each other in vivo. However, the role of SNS as a formula for obesity and the underlying mechanisms remains poorly understood.

Lipolysis is an important pathway of lipid metabolism. It hydrolyzes triacylglycerol (TAG) to produce glycerol and nonesterified free fatty acids (NEFA) (Arner, 2005). Lipolysis stimulation may contribute to weight loss and maintenance by increasing lipid turnover and promoting consumption (Fruhbeck et al., 2014). AMPK is considered to act as a key sensor of cellular energy level, which mediates phosphorylation and activation of lipolytic enzymes and exerts anti-lipolytic effects in adipocytes (Djouder et al., 2010; Wang et al., 2011). Research showed that the key step in AMP-activated protein kinase (AMPK) activation is the phosphorylation of the critical activation loop threonine 172 (Thr172) in the α-subunit (Carling et al., 2008). Anti-protein kinase B (AKT) modulates energy homeostasis by maintaining the level of ATP in cells and is a key negative regulator of AMPK activity (Hahn-Windgassen et al., 2005). It has been reported that decreased expression of p-AKT (Ser473) can lead to increased levels of p-AMPK (Thr172), thereby inhibiting lipolysis, and boosting Lipid droplet levels (Daval et al., 2005; Zhang et al., 2018).

In this study, the aim is to provide basic data to support the exploration of the pharmacological effects and potential mechanisms of SNS for obesity. We evaluated the effect of SNS on HFD-induced obesity in mice. Then, network pharmacology was used to elucidate plausible molecular targets of SNS and illuminate potential SNS-targeted signaling pathways.

2. Material and methods

2.1. Materials

Fetal bovine serum (FBS) and Dulbecco's modified Eagle's medium (DMEM) were purchased from Gibco Life Technologies (Gaithersburg, MD, USA), 3-isobutyl-1-methyl-7H-xanthine (IBMX, I7018), dexamethasone (D1756), rosiglitazone (R2408) and gelatin (48722) were obtained from Sigma (St. Louis, MO, USA). Insulin was purchased from Sanofi-Aventis Deutschland Gmbh (Frankfurt am Main, Germany). Methanol was obtained from Sinopharm Chemical Reagent Co., Ltd (Shanghai, China). 4% paraformaldehyde was purchased from Biosharp (Hefei, China). Paeoniflorin (Lot: No. M28GVB143089) and Saikosaponin A (Lot: No. P03M9F50593) were obtained from Shanghai Yuanye Bio-TechnologyCo., Ltd (Shanghai, China). Glycyrrhizic acid (Lot: No. DSTDG000601) was purchased from Lemeitian Pharmaceutical Technology Co., Ltd. (Chengdu, China). Naringin (Lot: No. 110722–200610) was obtained from National Institutes for Food and Drug Control (Beijing, China). HRP conjugated goat anti-rabbit IgG (H + L) was obtained from Servicebio (Wuhan, China). Anti-AMPKα, anti-phospho-AMPKα (Thr172), anti-AKT, anti-phospho-AKT (Ser473), anti-hormone-sensitive triglyceride lipase (HSL), anti-phospho-HSL (Ser660), and anti-adipose triglyceride lipase (ATGL) were purchased from Cell Signaling Technology Inc. (Danvers, MA, USA). Anti-phospho-ATGL (Ser406) was obtained from Abcam (Cambridge, MA, USA). Anti-beta tubulin was purchased from Proteintech Group (Rosemont, IL, USA).

2.2. Preparation of SNS

All consisting herbs of SNS that had passed internal quality control administrated by the National Medical Products Administration (China), were purchased from Shandong Baiweitang Chinese Herbal Medicine Drinks Slice Co. Ltd (China) and identified by Prof. Qingmei Guo from the Department of Pharmacognosy (Shandong University of Traditional Chinese medicine, China), and the full plant names have been checked via The Plant List (www.theplantlist.org) on August 20, 2022. The voucher specimens were deposited at our laboratory of Institute of Basic Medicine, the Second Hospital of Shandong University (Jinan, China). CH, BS, ZS, and GC were accurately weighed according to the ratio of 1:1:1:1. All herbal products were soaked in 10-fold volumes of distilled water for 30 min and boiled for 1 h using the condensation reflux method two times. The two aqueous extracts were pooled and concentrated to 1.248 g crude drug/mL by rotary evaporator. Prepared SNS were filtered with 100 μm mesh for the animal experiments and followed by sterilization using a membrane filter (0.22 μm) for the in vitro study. They were aliquoted and stored at −20°C until use.

2.3. HPLC analysis of SNS

High performance liquid chromatography (HPLC) analysis was carried on an Agilent 1260 Infinity II (Agilent Technologies Inc., CA, USA). The separation of SNS was performed on Agilent ZORBAX SB-C18 column (4.6 × 250 mm, 5 μm) at 30°C. The mobile phase consisted of 0.05% phosphoric acid (A) and acetonitrile (B) with a flow rate set at 1.0 mL/min. The gradient elution program was performed as Table 1. The injection volume was 10.00 μl and the detection wavelength was 210 nm with a diode array detector (DAD).

For preparation of the SNS sample, the prepared SNS was concentrated and dried with a rotary evaporator, then methanol was added, ultrasonicated for 40 min, and 0.416 g/mL solution was obtained by filtration. When preparing the reference solution, paeoniflorin, glycyrrhizic acid, saikosaponin A, and naringin were accurately weighed (Yan et al., 2012), and methanol was added to make the solution containing 0.5 mg per milliliter.

2.4. Animal study

Fifty male C57BL/6J mice (6 weeks old, weighing 15–17 g) were purchased from Beijing Vital River Laboratory Animal Technology Co., Ltd. (Beijing, China, License No. SCXK (Jing) 2021-0006), and housed in a barrier system maintained with controlled temperature (22 ± 1°C) and humidity (50–60%) under 12 h light/dark cycle. Mice were allowed free access to sterile water supplies and standard chow. After 1-week acclimation, mice were fed with HFD (60% Kcal from fat, D12492) except for the control group (negative control diet, 10% Kcal from fat, D12450B) (Research Diets, New Brunswick, NJ, USA). After 7 weeks, HFD-induced obese mice were randomly divided into four groups according to body weight, ten mice per group. For 4 weeks, mice were administrated with sterile water for the control and HFD group. According to the equivalent dose conversion between mice and human body, the daily dosage of SNS for adults (70 kg) was 24g. The designed gavage volume of mice was 0.1 mL per 10g body weight. The intragastric concentration of mice was calculated as 0.312 g/mL (SNS-L group). In addition, according to the Chinese medicine pharmacology animal dose gradient design principle, SNS-M group (0.624 g/mL) and SNS-H group (1.248 g/mL) were set up.

Fat mass and lean mass were measured by Bruker Minispec LF50 (Bruker, Germany). All animal experiments were approved by Research Ethics Committee of the Second Hospital of Shandong University (KYLL-2022LW119) and conducted according to Guiding Opinions on the Treatment of Laboratory Animals issued (Guoke Fa CAI Zi [2006] No. 398) by Ministry of Science and Technology of China.

Table 1
The gradient elution program.

Time (min)	0.05% phosphoric acid	acetonitrile
0.00	90.0	10.0
10.00	90.0	10.0
30.00	85.0	15.0
35.00	82.0	18.0
40.00	82.0	18.0
42.00	80.0	20.0
62.00	72.0	28.0
70.00	72.0	28.0
75.00	70.0	30.0
80.00	67.0	33.0
85.00	67.0	33.0
95.00	50.0	50.0
105.00	50.0	50.0
125.00	47.0	53.0
130.00	30.0	70.0
140.00	30.0	70.0
145.00	29.0	71.0
150.00	10.0	90.0

2.5. Histopathological examination

Mice adipose tissues were fixed in 4% paraformaldehyde for 5 days, dehydrated by ethyl alcohol, embedded in paraffin, and stained with hematoxylin and eosin (H&E) for histopathological assessment. All sections were observed by NanoZoomer S60a (Hamamatsu, Japan) to examine the morphology.

2.6. Isolation, culture, and treatment of primary mouse embryonic fibroblasts

Primary mouse embryonic fibroblasts (MEFs) were harvested by Embryos (12.5 to 13.5 days) as previously described (Djouder et al., 2010; Van Meter et al., 2014). After being seeded in different dishes that were pre-coated with collagen, MEFs were cultured with complete medium (CM) (4.5 mM glucose, 10% FBS, and 10% penicillin-streptomycin) for further experiments. MEFs were grown until 2 days post confluence. Adipogenesis was then induced using CM and a differentiation cocktail containing 5 μg/mL insulin, 1 μM rosiglitazone, 1 μM dexamethasone, and 0.5 mM IBMX for 4 days, after which media was changed to CM and 5 μg/mL insulin. After 2 days, cells were treated with CM and SNS (4 mg/mL) or CM for 24 h. The medium was changed every other day. MEFs were treated with different concentrations of SNS (2, 4, 8, 16 mg/mL) for 24 h, then determined Cell viability by Cell Counting Kit-8 (CCK-8) (Vazyme, Nanjing, China). Oil-red powder (Sigma, St. Louis, MO, USA) was dissolved in isopropanol solution (0.5 g/mL). Differentiated cells were fixed in 4% paraformaldehyde for 30 min and incubated in 60% isopropanol for 2 min. Samples are then stained in freshly diluted 60% Oil Red O for 20 min at 37°C, followed by 3 rinses in phosphate buffered saline. Images were captured by CKX53 (OLYMPUS, Tokyo, Japan). For quantification, the stained samples were incubated in 100% isopropyl alcohol for 15 min at room temperature to extract oil red O, and the absorbance was measured at 510 nm.

2.7. Measurement of TG and NEFA contents

Mouse serum was collected by centrifugation. Serum triglyceride levels were measured using a Triglyceride assay Kit (A110-1-1). NEFA in serum of mice or medium of MEF mature adipocytes were determined using a nonesterified free fatty acids assay kit (A042-2-1) following the manufacturer's instructions. All assay kits were obtained from Nanjing Jiancheng Bioengineering Institute (Nanjing, China).

2.8. Western blotting analysis

For Western blot analysis, cell samples and mouse WAT were captured and lysed in ice-cold RIPA buffer, respectively. Protein concentrations were measured by BCA Protein Quantification kit (Vazyme, Nanjing, China). Proteins (30 μg) were separated by SDS polyacrylamide gel and subsequently transferred onto PVDF membranes. After blocking in 5% nonfat milk solution in TBST for 1h, the membranes were incubated overnight at 4°Cwith primary antibody, then incubated with secondary antibody at room temperature for 1h. Protein bands were detected by ECL assay kit (Vazyme, Nanjing, China) and shown in fully automated chemiluminescence image analysis system (Tanon 5200 Multi, China).

2.9. Collecting formula-disease common targets

The therapeutic and mechanism targets of obesity were collected from The Comparative Toxicogenomics Database (CTD, http://ctdbase.org/) and GeneCards (https://www.genecards.org/) using "obesity" as index words. All the collected targets are overlapped further as the final target for the disease. The active compounds of every drug in SNS were detected from traditional medicine system pharmacology technology platform (TCMSP) database system (https://tcmsp-e.com/) (Ru et al., 2014) and Swiss Target Prediction (http://www.swisstargetprediction.ch/) with the standard of OB (oral bioavailability)≥ 30% and DL (drug-likeness)≥ 0.18. The primary active components documented in Pharmacopoeia of the People's Republic of China (ChP) or proved an intimate relationship with obesity would also be included. Then, the targets of these components were retrieved from this database. Uniprot (https://www.uniprot.org/) was used to convert all potential targets into standard Homo sapiens gene names.

2.10. Construction and analysis of PPI network

Putative SNS target-known therapeutic and mechanism targets of obesity network and putative SNS active compounds-target networks were constructed. All the target genes were imported into the STRING database (https://cn.string-db.org/), a database that enables visualization of all protein interactions and evaluation of interactions information to create a PPI network. The minimum interaction score required for this organism, which is limited to "Homo sapiens", is set to "Highest Confidence (>0.400)". All files will then be imported into Cytoscape3.8.2 for further analysis.

2.11. GO and KEGG pathway enrichment analysis

The biological process, molecular function, the pathway of Gene Ontology (GO) functional enrichment analysis, and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis were gathered using Metascape (https://metascape.org/) (Zhou et al., 2019). Then, bar and bubble charts were drawn.

2.12. Statistical analysis

Biostatistical analyses were presented as mean ± SD (standard

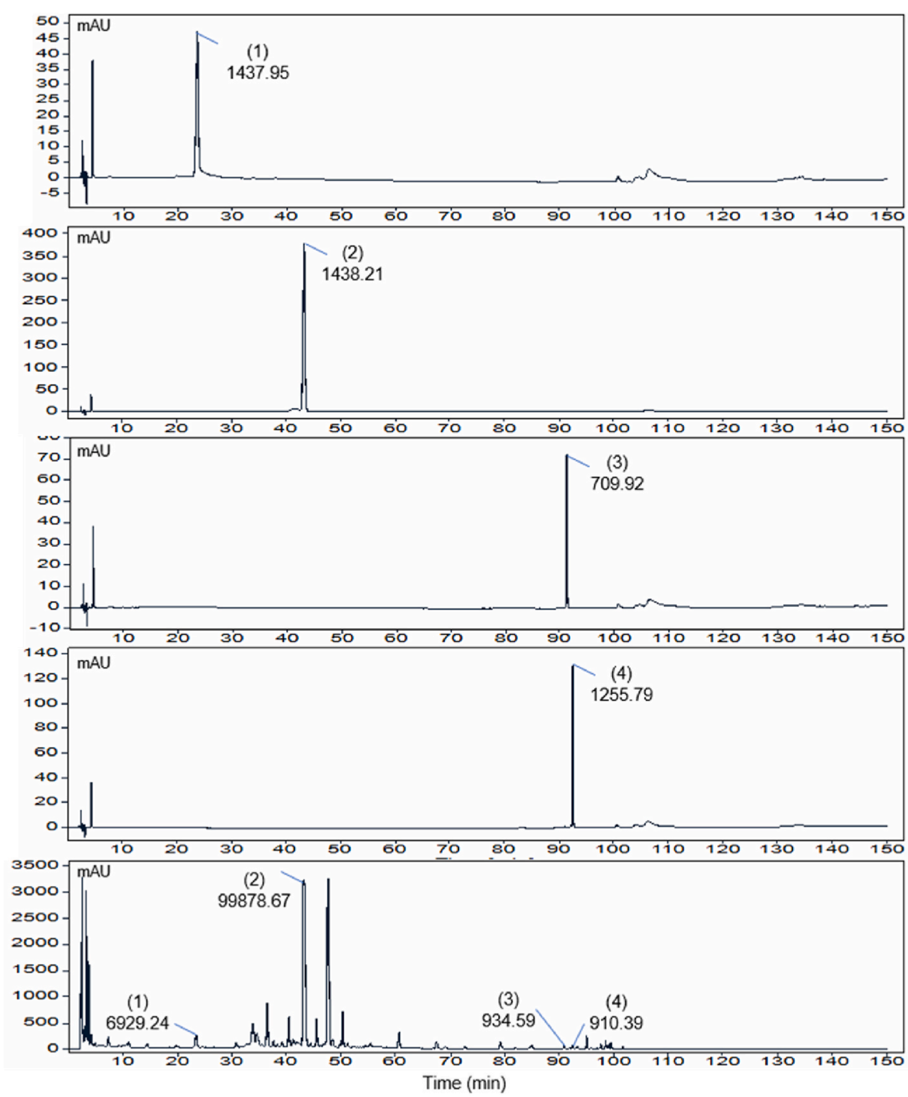

Fig. 1. The HPLC chromatogram of SNS. Standard chromatogram: (1) paeoniflorin; (2) naringin; (3) glycyrrhizic acid; (4) saikosaponin A; and SNS. The peak label represents the peak area.

deviation) by the GraphPad Prism 8 software package (GraphPad, La Jolla, CA, USA). Results are representative of at least three independent experiments or at least five mice for each group. To assess statistical significance, we performed an unpaired two-tailed Student's t-test for comparison of data distributed in a normal pattern between two groups. $P \leq 0.05$ was considered statistically significant.

3. Results

3.1. The HPLC chromatogram of SNS

HPLC was used to calibrate four active components in SNS, including (1) paeoniflorin, (2) naringin, (3) glycyrrhizic acid, and (4) saikosaponin A. Chromatographic peaks were identified according to UV–visible spectroscopy and retention time. The peak height and peak area of each component were given in Fig. 1.

3.2. Effects of SNS on body weight and fat mass changes

To assess the repercussions of SNS on the development of obesity, mice were fed an HFD diet for 7 weeks to induce obesity, then by administration with three different doses of SNS (0.312, 0.624, and 1.248 g/mL) for additional 4 weeks with continuous HFD diet (Fig. 2A). From week six, the body weight in the HFD group was remarkably higher than that in the control group. SNS-M and SNS-H significantly reduced the HFD-induced body weight gain without affecting food intake (Fig. 2B and C). In addition, Nuclear Magnetic Resonance analysis demonstrated that the fat mass was markedly decreased in the SNS groups after four weeks of administration, and the lean mass content increased significantly (Fig. 2D, E, and F). These data suggested that SNS significantly reduced fat accumulation and increased lean meat content in HFD-induced obesity mice without affecting the amount of average food intake. In line with the above findings, serum biochemistry assays then demonstrated that SNS significantly decreased the serum TG and NEFA levels in obesity mice (Fig. 2G and H). These results indicated that SNS exerted beneficial effects on the treatment of obesity and avoided the effect of lipotoxicity on the body to a certain extent.

3.3. Effects of SNS on histological changes of adipose tissues

Adipose tissue expansion is one of the main manifestations of obesity (Vishvanath and Gupta, 2019). After mice were sacrificed and epididymal adipose was collected, we found that SNS could make mice thinner (Fig. 3A) and reduce the volume of epididymal adipose (Fig. 3B) compared with HFD group. To determine the histological changes of adipocytes, we performed H&E staining and quantified adipocyte size in WAT, showing that SNS induced a significant decrease in the percentage of big-sized adipocytes in obese mice. Compared with the HFD group, the average size of adipocytes was dramatically smaller in the SNS group (Fig. 3C). These results suggested that SNS prevents HFD-induced adipocyte hypertrophy.

3.4. Prediction for bioactive compounds of SNS and potential targets against obesity

To discover the potential pharmaco-mechanisms of SNS in the treatment of obesity, under the screening condition of OB $\geq$ 30% and DL $\geq$ 0.18, the present study screened out 135 candidate active compounds and their 534 potential target genes of four herbs in SNS based on TCMSP, Chinese Pharmacopoeia, and Swiss Target Prediction. Then, the SNS-component-target network was constructed by Cytoscape 3.7.2, and the size of the node was determined by the degree value (Fig. 4A).

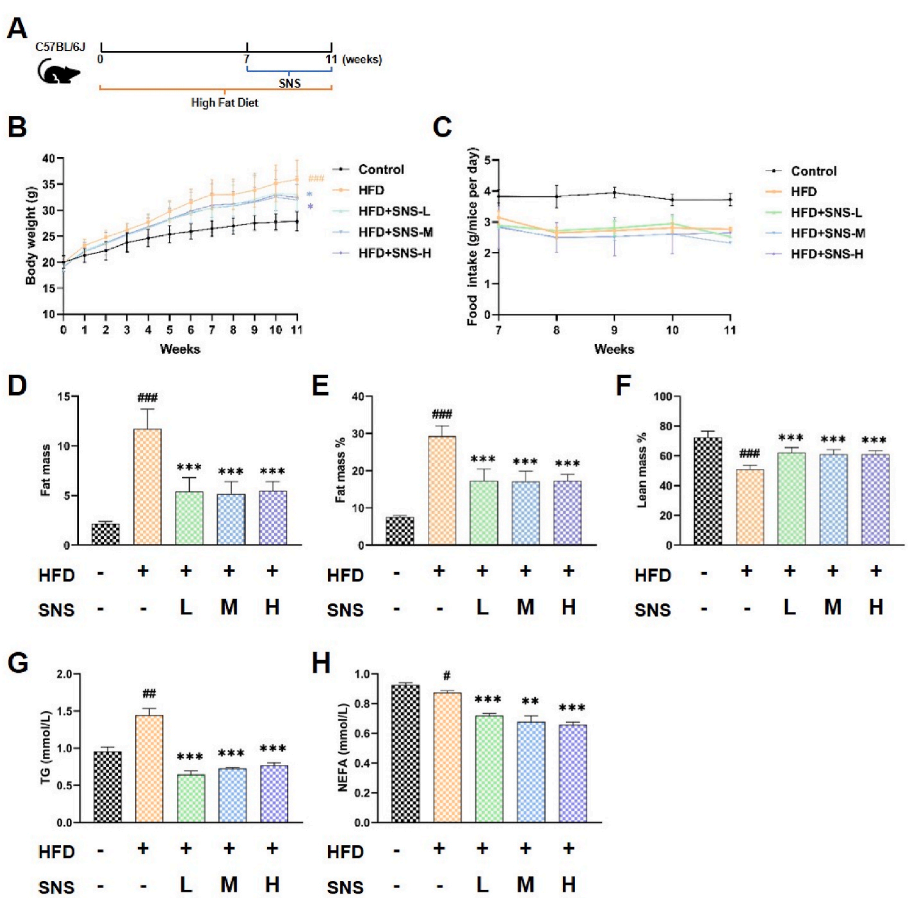

Fig. 2. Effects of SNS on body weight and fat mass changes. (A) HFD-induced obesity in mice was built as described in the method, different doses of SNS were given from week 8 to week 11. (B) The body weight of mice was regularly measured weekly ($n = 10$). (C) Food intake was measured weekly after SNS was administrated. fat mass (D), fat mass rate (E), and lean mass rate (F) were measured and calculated at the end of the animal experiment. (G, H) Content of TG and NEFA in serum with or without treatment of SNS. Data are represented as the mean $\pm$ SD. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$ vs. HFD group, #$p < 0.05$, ##$p < 0.01$, ###$p < 0.001$ vs. control group ($n = 5$).

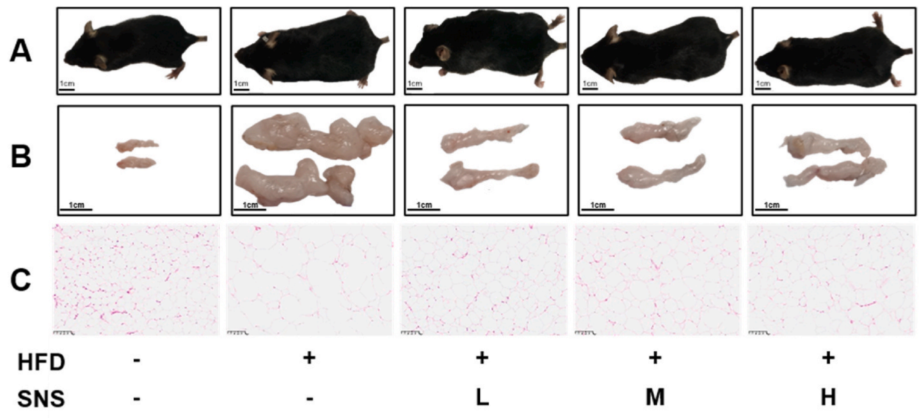

Fig. 3. Effects of SNS on histological changes of adipose tissues. (A) Body condition. (B) Epididymal adipose. Scale bar = 1 cm. (C) Representative images of H&E staining of epididymal adipose tissues in each group. Scale bar = 100 μm.

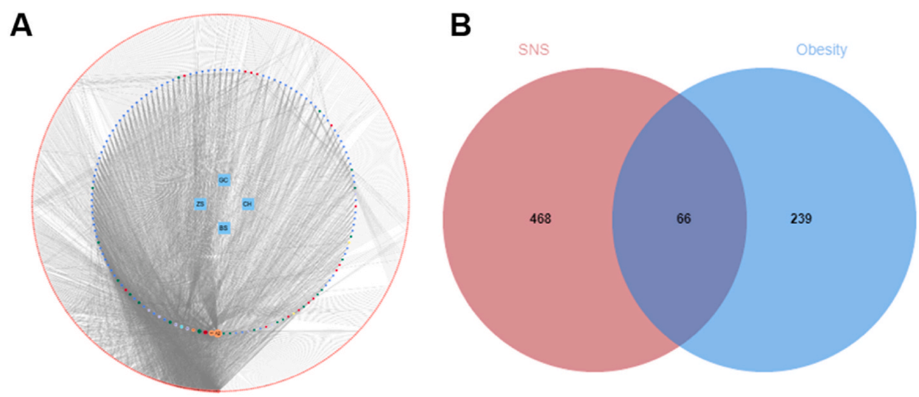

Fig. 4. Prediction for bioactive compounds of SNS and potential targets against obesity. (A) SNS-component-target network. Four points in the center represent four herbs in SNS, the points in the inner circle represent compounds and the points in the outer circle potential targets. (B) Venn diagram of the target genes in SNS and obesity.

The first ten potential bioactive compounds are quercetin, kaempferol, Saikosaponin D, Obacunone, isorhamnetin, albiflorin_qt, naringenin, paeoniflorin, luteolin, and 7-Methoxy-2-methyl isoflavone as shown in Table 2. Furthermore, 305 therapeutic targets for obesity were found through Genecards and CTD. Next, the targets of SNS and obesity were used to create a Venn diagram (Fig. 4B), and 66 potential therapeutic targets for SNS against obesity were eventually collected.

3.5. Construction of PPI network

The potential therapeutic targets for SNS acting on obesity were imported into the STRING database for protein-protein interaction analysis. A PPI network with 64 nodes and 547 edges was obtained. The higher degree value determines the bigger size and darker red color of the node. The result shows that IL6, TNF, AKT1, PPARG, TP53, PPARA, ADIPOQ, PTGS2, SIRT1, and ESR1 interact with other nodes closely, and may be the key proteins in treating obesity (Fig. 5A).

Table 2
The first ten potential active compounds ranked by degree and OB ≥ 30% or DL ≥ 0.18.

ID	Molecule ID	Potential active ingredients	Degree	MW	OB (%)	DL
A2	MOL000098	quercetin	310	302.25	46.43	0.28
A1	MOL000422	kaempferol	188	286.25	41.88	0.24
CH10	MOL004637	Saikosaponin D	102	781.1	34.39	0.09
ZS14	MOL013352	Obacunone	101	454.56	43.29	0.77
A3	MOL000354	isorhamnetin	76	316.28	49.6	0.31
BS6	MOL001928	albiflorin_qt	74	318.35	66.64	0.33
C1	MOL004328	naringenin	74	272.27	59.29	0.21
BS5	MOL001924	paeoniflorin	68	480.51	53.87	0.79
ZS1	MOL000006	luteolin	56	286.25	36.16	0.25
GC12	MOL003896	7-Methoxy-2-methyl isoflavone	44	266.31	42.56	0.2

3.6. GO and KEGG enrichment analysis

To further illustrate the main biological functions of SNS-obesity putative targets, GO and KEGG pathway analyses were performed. Biological Process (BP), Cellular Component (CC), and Molecular Function (MF) had 968, 33, and 74 terms, respectively. Based on the P value, the top 10 BP, CC, and MP of the Go enrichment analysis were selected for visualization (Fig. 5B). The results showed that regulation of small molecule metabolic process, regulation of lipid metabolic process, membrane raft, membrane microdomain, nuclear receptor activity, and lipid binding were the important biological functions. These functional terms with high response values have been widely concerned in the study of lipid metabolism. Additionally, a total of 108 signaling pathways were obtained from KEGG enrichment analysis. According to p-value (Fig. 5C), we selected the top 20 pathways for research. The KEGG enrichment data showed important factors of SNS in the treatment of obesity. It includes Lipid and atherosclerosis, Insulin resistance, Type II diabetes mellitus, Insulin signaling pathway, Non-alcoholic fatty liver disease, AMPK signaling pathway, and regulation of lipolysis in adipocytes. It is interesting to note that multiple biological functions and

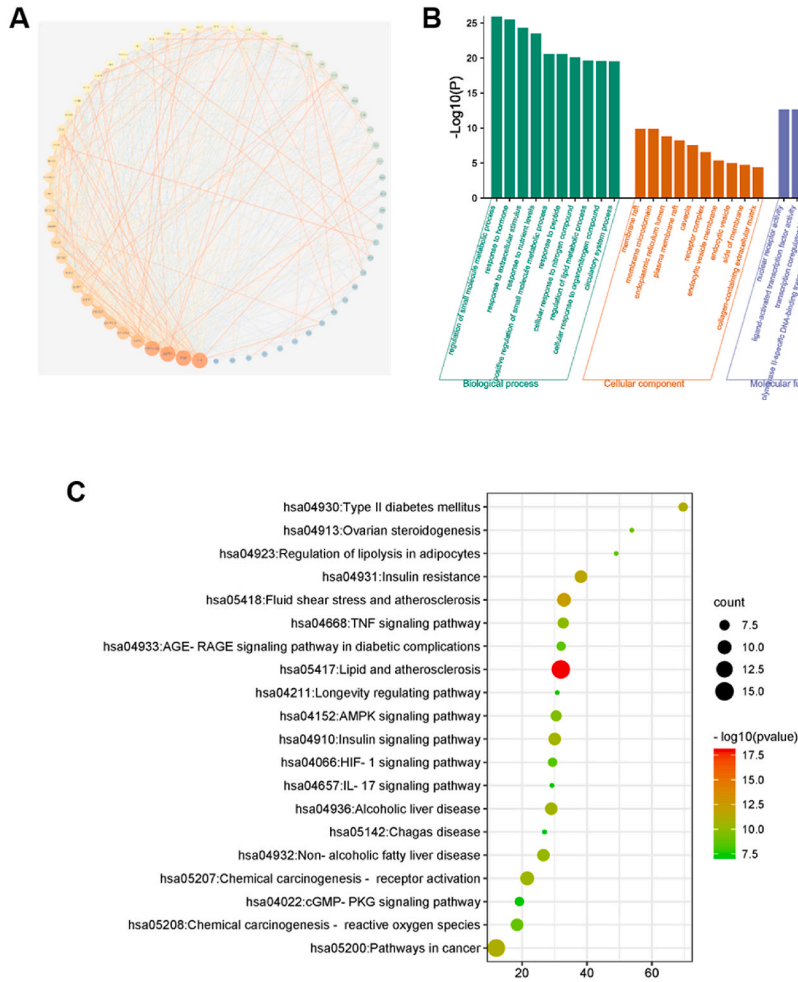

Fig. 5. GO and KEGG enrichment analysis. (A) PPI network for SNS treating obesity. Bubble charts for GO pathway enrichment analysis(B) and KEGG pathway enrichment analysis(C).

signaling pathways are closely linked with lipolysis. The current study demonstrates that Inhibition of AMPK protein expression significantly reduced the weight of adipose tissue and the size of adipocytes (Daval et al., 2005), which is consistent with our experimental results of SNS promoting lipolysis and alleviating obesity in the mice model. In the following study, we focused on the regulation of lipolysis in adipocytes and AMPK signaling pathway.

3.7. Effect of SNS on the activity of the protein in WAT of mice

To investigate whether SNS reduced lipid accumulation in obese mice by affecting lipolysis, we detected the critical targets of AMPK signaling pathway via western blotting. As shown in Fig. 6A and B, the protein phosphorylation levels of AKT, HSL, and ATGL were significantly increased, and the phosphorylation levels of AMPKα protein were significantly reduced in the adipose tissue of mice in each dose group of SNS, relative to HFD group. These results indicate that SNS can suppress WAT lipolysis by inhibiting AMPK activity, the AKT/AMPK/HSL signaling pathway.

3.8. Effect of SNS on lipid accumulation in MEFs

Given the lipid-lowering effects of SNS in vivo environment, we isolated MEFs and differentiated them into adipocytes to further examine whether SNS directly alleviates lipid accumulation. In cell therapy experiments, the doses of SNS were selected according to our CCK-8 data. The results showed that SNS inhibited the viability of MEFs cells in a dose-dependent fashion. Because a 24 h incubation as no significant difference in cell viability at a concentration of 4 mg/mL compared with control (Fig. 7A), we chose this concentration for subsequent experiments. As shown in Fig. 7B, compared with the control group, SNS-treated adipocytes showed greater fatty acid secretion in medium. The results of oil Red O experiment showed that SNS treatment significantly reduced the number of lipid droplets and lipid content in MEFs (Fig. 7C and D). Interestingly, compared with the control group, lipid droplets in the SNS group mainly were dispersed in the cytoplasm in smaller sizes, which further verified the role of SNS in lipolysis. The Western blot analyses indicated that SNS significantly decreased the expression and blunted the activation of AMPK. Consistent with a reduction in the p-AMPKα (Thr172) level, increase in phosphorylation of HSL and ATGL (Fig. 8A and B). The above results are consistent with the results in vivo.

4. Discussion

Obesity, a chronic relapsing progressive disease process (Bray et al., 2017), has become a worldwide public health problem with a continuously increasing prevalence (Lin and Li, 2021). Since obese patients often suffer from complications, making safety assessments of AOMs are more complex and expensive to develop (Muller et al., 2022). SNS is a prestigious and traditional prescription used to treat a variety of digestive disorders. In the present study, we found that SNS significantly reduced fat accumulation in both HFD-induced obesity mouse model and adipocyte model in vitro, due to increased lipolysis. SNS

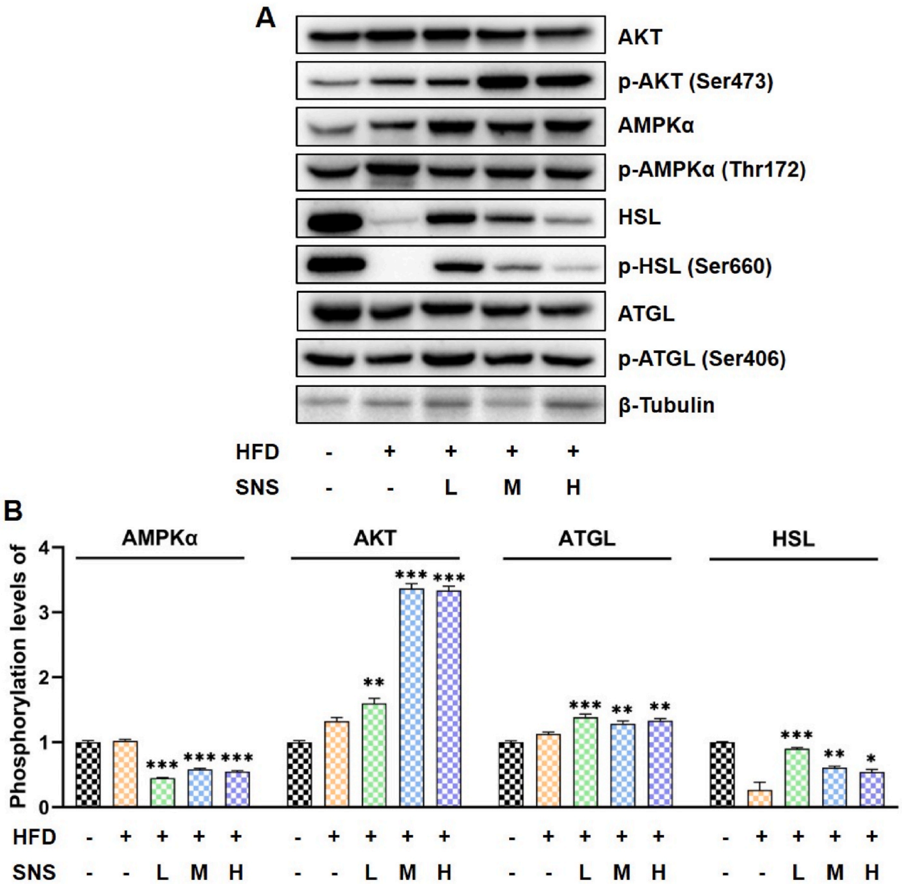

Fig. 6. Effect of SNS on the activity of the protein in WAT of mice. (A) The protein expression in epididymal adipose tissue of AKT, p-AKT (Ser473), AMPK, p-AMPKα (Thr172), HSL, p-HSL (Ser660), ATGL, p-ATGL (Ser406) were determined by Western blot analysis. (B) The representative blot images are shown and the ratios of p-AKT (Ser473)/AKT, p-AMPKα (Thr172)/APMKα, p-HSL (Ser660)/HSL, and p-ATGL (Ser406)/ATGL are calculated. The results are expressed as the means ± SD from three independent experiments. *p < 0.05, **p < 0.01, ***p < 0.001 vs. HFD group (n = 3).

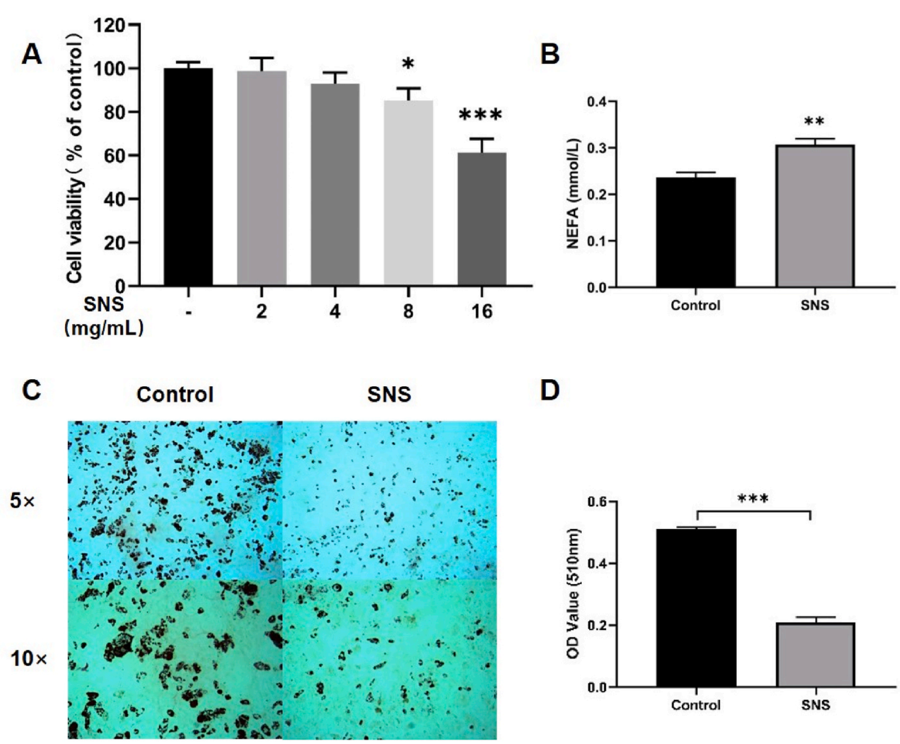

Fig. 7. Effect of SNS on Lipid Accumulation in MEFs. MEF cells were treated with various concentrations of SNS (2,4,8,16 mg/mL) for 24 h. (A) Cell availability was determined by CCK-8 assay. (B) The content of NEFA in the medium of differentiated MEF adipocytes was determined with or without SNS. (C) Representative images of MEF cells stained with Oil Red O staining were photographed (magnification, 50 × and 100×), and (D) Oil Red O in cells were collected and measured. The results are expressed as the means ± SD from three independent experiments. *p < 0.05, **p < 0.01, ***p < 0.001 vs. control (n = 3). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

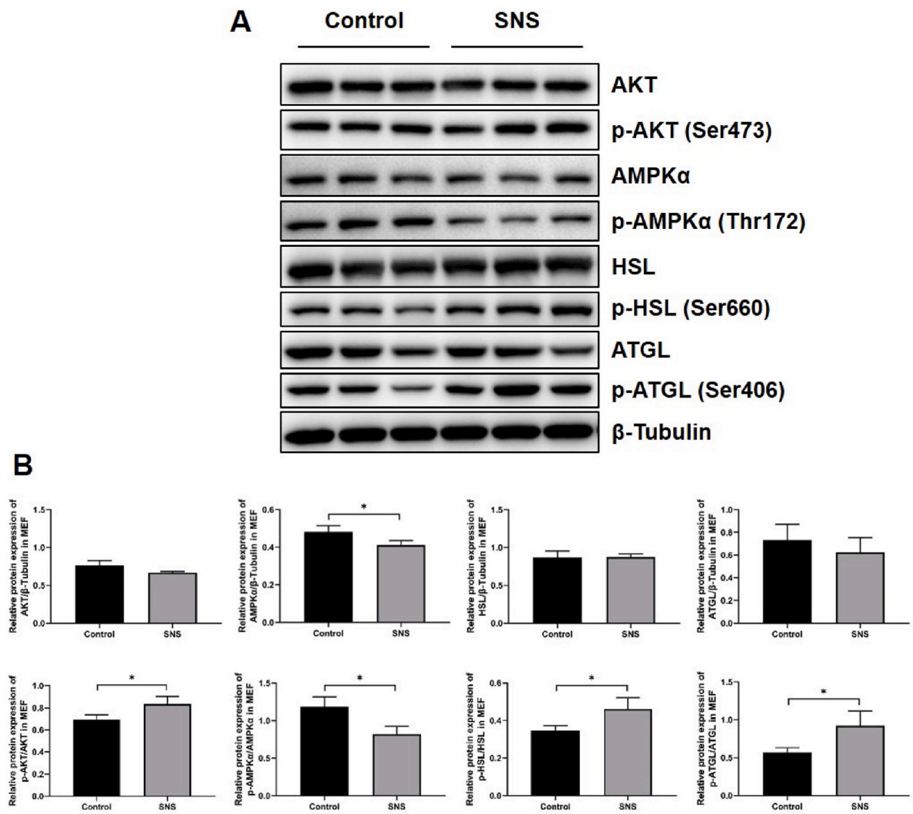

Fig. 8. Effect of SNS on Lipid Accumulation in MEFs. MEF cells were treated with 4 mg/mL SNS for 24 h. (A) The protein expression of AKT, p-AKT (Ser473), AMPK, p-AMPKα (Thr172), HSL, p-HSL (Ser660), ATGL, p-ATGL (Ser406) were determined by Western blot analysis. (B)The representative blot images are shown and the ratios of p-AKT (Ser473)/AKT, p-AMPKα (Thr172)/APMKα, p-HSL (Ser660)/HSL, and p-ATGL (Ser406)/ATGL are calculated. The results are expressed as the means ± SD from three independent experiments. *p < 0.05 vs. control (n = 3).

significantly activated lipolysis of adipocytes through phosphorylation of AKT and reverse regulate AMPK and following activation of HSL, thereby promoting the phosphorylation of ATGL and improving lipid metabolism.

In order to control the quality of SNS, we successfully identified four monomeric components from four medicinal materials in the prescription by HPLC, which play an important role in regulating lipid metabolism. Saikosaponin A, a major kind of ingredient in CH, has been proven a significant role in the fatty acid biosynthesis by down-regulating Fasn and Acaca (Li et al., 2021). Naringin, which is mainly derived from ZS, reduced fatty acid uptake and denovo lipogenesis through CD36/ACC pathway and enhanced fatty acid oxidation through PPARα/CPT-1 pathway (Zhang, X. et al., 2021b). Paeoniflorin, a major component of BS, has improved fructose-induced insulin resistance and hepatic steatosis by activating LKB1/AMPK and AKT pathways (Li et al., 2018). As for the glycyrrhizic acid, derived from GC, it can suppress early stage of adipogenesis in 3T3-L1 cells via repression of MEK/ERK-mediated C/EBPβ and C/EBPδ expression (Yamamoto et al., 2021). These results suggest that SNS can improve obesity through a variety of pathways.

SNS was observed with a significant role in decreasing serum TG in clinic. It was also confirmed in numerous experiments that SNS had a significant decreasing effect on total cholesterol and TG content in mice fed with a HFD (Wei et al., 2021). This result was confirmed by our experiments. Previous reports have explored the role of SNS in regulating immune function. However, it does not talk about the role of its direct lipid-lowering effect and deeper mechanism in anti-obesity (Wang et al., 2021; Zong et al., 2019). In the present study, SNS reduced body weight in obese mice with no effect on food intake. Further tests showed that the body fat content of mice in the treatment group was significantly decreased and the lean mass was significantly increased, indicating that SNS efficiently improves lipid metabolism in obese mice. At the same time, pathological examination showed that the percentage of large adipocytes decreased significantly during SNS treatment. These results suggest that part of the mechanism of SNS reducing obesity in mice is directly acting on adipose tissue, affecting the formation or decomposition of lipid droplets in adipose tissue, resulting the reduction of adipocyte volume. To test this conjecture, we performed a network pharmacology analysis.

Network pharmacology can systematically analyze the complex relationships among drugs, biological systems, and diseases and predict the underlying mechanisms of action by constructing interaction networks between drugs and proteins or genes and diseases (Hopkins, 2008). For the present study, we collected the potential components and targets of SNS from public databases. The top ten candidate active ingredients, particularly quercetin and Saikosaponin D, all regulate the stability of lipid metabolism to varying degrees. Quercetin induces the browning of subcutaneous WAT via the AMPK/sirtuin1/PGC1α pathway (Lee et al., 2017), which results in an increased flux of TG-derived fatty acid towards subcutaneous WAT, and is responsible for the improved plasma TG levels (Kuipers et al., 2018). It has been well characterized that Saikosaponin D prevented glycerolipid accumulation, inhibited fatty acid biosynthesis, promoted fatty acid degradation, and subsequently recovered hepatic lipid homeostasis (Gu et al., 2022; Li et al., 2021). Then, PPI network analysis showed that IL6, TNF, AKT1, PPARG, TP53, PPARA, and ADIPOQ might be the crucial proteins of SNS in treating obesity. A large amount of evidence has shown that chronic inflammation in adipose tissue may play a vital role in obesity-related metabolic dysfunction (Chakarov et al., 2022; Cottam et al., 2022; Liu et al., 2022), among which IL6 and TNF are important proinflammatory factors (Kumari et al., 2018), which makes previous studies mainly focus on anti-inflammatory mechanisms. Interestingly, AKT1 (Ding et al., 2017; Izumiya et al., 2008; Sanchez-Gurmaches et al., 2019), PPARs (Gross et al., 2017; Markussen et al., 2022), and ADIPOQ (Tanabe et al., 2015) played a critical role in maintaining lipid homeostasis. Furthermore, pathway and function enrichment analysis also showed that SNS could regulate pathways related to lipid metabolism, including the AMPK signaling pathway, regulation of lipolysis in adipocytes, lipid and atherosclerosis, and insulin signaling pathway. These preliminary prediction results encourage us to investigate further the combat obesity effects of SNS in direct regulation of lipid metabolism.

In eukaryotes like mammals, AMPK is one of the central regulators coordinating metabolism (Mihaylova and Shaw, 2011) and specialized roles in the regulation of TG turnover in adipose tissues (Trefts and Shaw, 2021). As a key component in the controlling of whole-body energy balance, WAT has the unique function of storing TAG for hydrolysis to provide fuel for other organs during times of low carbohydrate availability (Rosen and Spiegelman, 2006). Lipolysis is accomplished through sequential hydrolysis of three fatty acids residues of TAG catalyzed by lipase. ATGL is considered to be the rate-limiting enzyme that initiates fat mobilization and performs the first step in TAG catabolism to produce diacylglycerol, which is then degraded by HSL to monoacylglycerol and finally to glycerol and fatty acids by MGL (Zechner et al., 2017). Previous studies have demonstrated that AMPK can negatively regulate HSL activity and inhibit lipolysis under stimulated conditions (Djouder et al., 2010; Kim et al., 2016; Steinberg and Kemp, 2009). In our study, we demonstrated that SNS markedly decreased the phosphorylation of AMPK and increased the phosphorylation of HSL and ATGL. It has previously been reported that increased levels of phosphorylated AKT negatively regulate AMPK activity (Hahn-Windgassen et al., 2005; Zhao et al., 2017), and reduce lipid droplet accumulation (Zhang et al., 2018). Here, we observed significant increase in levels of p-AKT in adipose tissue challenged with different doses of SNS. Our study provides critical evidence that SNS promoted lipolysis activity and improved lipid metabolism through the AKT/AMPK/HSL axis.

However, this study has some limitations. It is reported that excessive lipolytic rate leads to the release of excess free fatty acid, which may become harmful to other tissues or organs (especially liver) (Wang et al., 2011). Interestingly, although the secretion of NEFA in the medium of MEF adipocytes was significantly increased in the SNS group in vitro, the content of NEFA in the serum of mice was significantly decreased. KyeongJin Kim thinks the difference is related to fatty acid oxidation activity in brown adipose tissue and liver (Kim et al., 2021). Numerous studies have shown that SNS has a protective effect on the liver (Xu et al., 2022). This study did not clarify how SNS can promote lipolysis and protect liver against lipotoxicity at the same time, which will be further explored in the following study.

5. Conclusion

In this study, the HFD-induced obesity mouse model confirmed that SNS exhibits a significant lipid-lowering effect. Subsequently, the mechanism of SNS against obesity was generated to explore the underlying through network pharmacology. The results of pharmacological experiments showed that the mechanism of SNS in obesity might be related to the activation of AMPK signaling pathway to promote lipolysis. Our study offers vital evidence for the clinical use of SNS in the treatment of obesity.

Funding

This research was supported and funded by the National Natural Science Foundation of China (81773997) and the Shandong Province Key Research Program (2016ZDJS07A21). Rong Sun is supported by research fund "Traditional Chinese medicine pharmacology and toxicology expert (ts201511107)" from the Taishan Scholar Project of Shandong Province (China).

CRediT authorship contribution statement

Jianchao Li: designed and did the major experimental work. Kaiyi Wu: participated in part of the experiments. Ying Zhong: contributed to manuscript writing. Jiangying Kuang: supervised the research. Nana Huang: supervised the research. Xin Guo: contributed to manuscript writing. Chong Guo: participated in part of the experiments. Hang Du: contributed to manuscript writing. Rongrong Li: participated in part of the experiments. Xiaomin Zhu: participated in part of the experiments. Tianyu Zhang: participated in part of the experiments. Liping Gong: contributed to manuscript writing. Lisong Sheng: contributed to manuscript writing. Rong Sun: designed and supervised the research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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